

A. Proofs

1) Proof of Proposition 1:

Proof Sketch. The sum of distances is easy to check in polynomial time, i.e., the problem is in NP. To prove the NP-Hardness, we show a reduction from the decision version of the Traveling Salesman Problem (TSP), one of Karp's 21 NP-Complete problems [49].

We first introduce the decision version of the TSP problem: Given an undirected weighted graph $G = (V, E)$ with n vertices and their pairwise distances, the problem is to decide whether there exists a path of total weight less than or equal to L , which visits each vertex exactly once. In addition, if all the vertices are in a plane (i.e., the distances satisfy the triangle inequality), existing studies also prove that the TSP problem is still NP-Complete under l_1 and l_2 distances [50]–[52].

Given an instance of the TSP problem in a plane, we then construct an instance of the Constant Pattern Ordering Problem with 2 dimensions: For each vertex $v_i \in V$, we use $(v_i[1], v_i[2])$ to denote its coordinates, and then construct a tuple $y_i = (v_i[1], v_i[2]) \in Y$ corresponding to v_i . Therefore, the distances between vertices v_i, v_j are also equal to the distances between tuples y_i, y_j .

Following Problem 2, the decision version of our problem is to decide if there exists a constant pattern order with the sum of the distances less than or equal to K . Let $K = L$, we prove that these two instances are equivalent. Note that the distances of two instances are equal, to find a path of total weight less than or equal to L in TSP problem, we then order the tuples following the order of the vertices in the path. Therefore, the path in TSP has total weight less than or equal to L , if and only if there exists an order of tuples with the sum of the distances less than or equal to K . To conclude, the NP-Completeness of the decision problem is proved. $\square$

2) Proof of Lemma 2:

Proof. First, we consider the distance between x and y in the j -th dimension. When $Y_i^{\min}[j] < x[j] < Y_i^{\max}[j]$, $x[j]$ is covered by the range of $Y_i[j]$. Thereby, $\theta_j(x, Y_i) = 0$ in this case. When $x[j] < Y_i^{\min}[j]$, $x[j]$ is out of the value range of the j -th dimension. Then the value closest to $x[j]$ is $Y_i^{\min}[j]$, which gives the minimum distance $|x[j] - Y_i^{\min}[j]|$. Similarly, we could get the distance bound of the j -th dimension when $x[j] > Y_i^{\max}[j]$. In summary, we have

$$|x[j] - y[j]| \geq \theta_j(x, Y_i), \quad \forall y \in Y_i, \quad 0 \leq j \leq l$$

The distance $\delta(x, y)$ is thus bounded by $\theta(x, Y_i)$, for $y \in Y_i$.

$$\delta(x, y) = \sum_{j=1}^l |x[j] - y[j]| \geq \sum_{j=1}^l \theta_j(x, Y_i) = \theta(x, Y_i)$$

$\square$

3) Proof of Proposition 3:

Proof. Suppose that there exists a neighbor y^* of x in cluster Y_i , i.e., $\delta(x, y^*) \leq \delta_\kappa$. On the other hand, according to the Lemma 2, $\forall y \in Y_i, \delta(x, y) \geq \theta(x, Y_i)$. It follows $\delta(x, y^*) \geq \theta(x, Y_i)$. We thus obtain $\theta(x, Y_i) \leq \delta_\kappa$. $\square$

4) Proof of Proposition 4:

Proof. The proposition can be proved similarly following the proof of Proposition 3. $\square$

5) Proof of Proposition 5:

Proof. Note that the smoothing tuples in our algorithm are selected from the valid modification set V_i . Thereby, the modification x'_i of x_i exists if and only if V_i is not empty. We discuss two cases of window W_i .

If $W_i = \emptyset$, the smoothness constraint $\mathcal{S}$ of x_i is naturally satisfied. Therefore, a candidate can always be found from $\text{Nei}(x_i)$ in constant patterns Y to modify x_i safely.

If $W_i \neq \emptyset$, then the candidate modification set C_i involves $\text{Nei}(x'_j)$, for $x'_j \in W_i$, i.e., $x'_j \in C_i$ as well. Due to the streaming computation, x_j is smoothed before x_i . That is, x'_j in the window satisfies the smoothness constraint with each other, referring to the previous smoothing steps. Therefore, there always exists $x'_j \in W_i$ as the valid modification of x_i . $\square$

6) Proof of Proposition 6:

Proof. For window size $\omega = 0$, the temporal smoothness of the time series is no longer considered. The window W_i of each tuple x_i is always empty and the candidate set C_i is $\text{Nei}(x_i)$. Since the modifications no longer affect each other, we have $V_i = C_i$. The one with the minimum distance to x_i in V_i returned by Algorithm 2 is indeed the optimal solution. $\square$

B. Application Evaluation

In this section, we share the real-world experiences of applying our proposal with constant patterns at our industrial partners and on a public benchmark. Time series prediction is the most common application, including fuel consumption prediction [3] and wind turbine icing forecasts. It is also crucial for T-BOX in classifying dump truck trajectories. However, data quality issues make predictions and classifications on raw data unsatisfactory.

Figure 17 shows these applications without/with different smoothing methods. For prediction, we use LSTM [53]. For classification, we use DTW [54] as the distance measure and 1NN [55] as the classifier. (1) For fuel consumption prediction, Tianyuan Inc. aims to forecast vehicle fuel use for efficiency purposes [3] (Section V-A1). Compared to other cleaning methods, our method achieves the lowest prediction RMSE, as shown in Figure 17a. (2) For dump truck trajectory classification, Weichai Power requires accurate trajectory categorization (Section V-A2). Raw data errors make direct classification unreliable, but applying CCFDSmooth significantly improves accuracy, reaching nearly 90% (Figure 17b). (3)

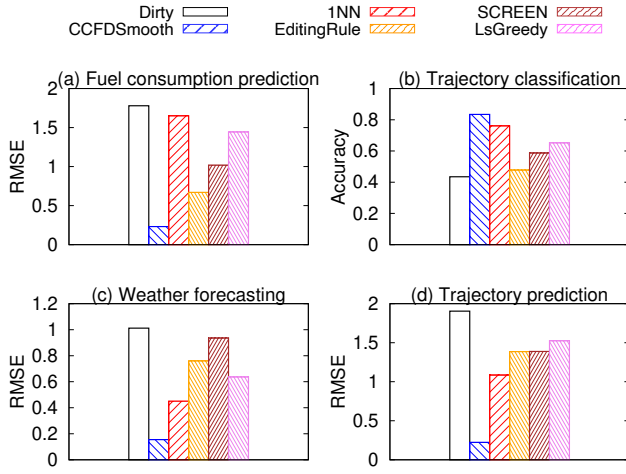


Fig. 17: Applications with or without data cleaning

For weather forecasting, Goldwind Co. predicts conditions for wind turbine icing early warnings (Section V-A3). Our method achieves an RMSE only half that of 1NN, thanks to superior cleaning performance (Figure 17c). (4) For taxi trajectory prediction over a public dataset [30] (Section V-B1), time series cleaning substantially enhances prediction performance, with CCFDSmooth achieving the lowest RMSE among all methods (Figure 17d).